

The formula first passes an array of three values to the LARGE function, and then uses the SUM function to add the values returned by the LARGE function. Notice that the values 1 through 3 are enclosed in brackets rather than parentheses. After typing an array formula, press [Ctrl]+[Shift]+[Enter] instead of [Enter].

Formulas of this type can become unwieldy as *n* gets larger. For example, to sum the top 30 values in a range, a formula must contain a list of integers from 1 to 30. Here is a more general version of the array formula:

```
=SUM(LARGE(A1:A100,ROW(INDIRECT('1:30'))))
```

This formula uses the ROW function to generate a series of integers between 1 and 30, and uses this array as the second argument for the LARGE function. To sum a different quantity of numbers, just change the 30 to the desired number.

3.6.1.6 Rounding To N Significant Digits

Excel includes three functions (ROUND, ROUNDUP, and ROUNDDOWN) that round values to a specified number of digits. In some cases, however, you may need to round a value to a specified number of significant digits.

For example, you might want to express the value 1,432,187 in terms of two significant digits (that is, as 1,400,000). Here's an elegant solution. The formula below rounds the value in cell A1 to the number of significant digits specified in cell A2:

```
=ROUND(A1,A2-1-INT(LOG10(ABS(A1))))
```

3.6.1.7 Calculate The Day Of The Year And Days Remaining

If you've ever had to figure out which of the year's 365 days a particular date falls on, or how many days remain in the year, you've probably found that Excel lacks functions to perform the calculation. But you can create formulas to do the job.